

UNIT 4 SERIES

11.1 Sequences

- A sequence

$$a_1, a_2, a_3, a_4, \dots$$

is an ordered arrangement of real numbers, one for each positive integer. We denote it by $\{a_n\}_{n=1}^{\infty}$ or simply $\{a_n\}$.

- A sequence may be specified by giving enough initial terms to establish a pattern, such as

$$1, 4, 7, 10, 13, \dots \quad \text{or} \quad \{1, 4, 7, 10, 13, \dots\}$$

by an **explicit formula** for the n th term, such as

$$a_n = 3n - 2, \quad n \geq 1 \quad \text{or} \quad \{3n - 2\}_{n=1}^{\infty}$$

or by a **recursive formula**

$$a_1 = 1, \quad a_n = a_{n-1} + 3, \quad n \geq 2$$

- **Example** List the first 4 terms of the following sequences:

(a) $a_n = \frac{n}{n+1}, \quad n \geq 1$

(b) $b_n = \frac{(-1)^n(n+1)}{3^n}$

(c) $\{\sqrt{n-3}\}_{n=3}^{\infty}$

(d) $\{0.999\}$

- **Example** Find a formula for the general term a_n of the sequence, assuming that the pattern of the first few terms continues.

$$\left\{ \frac{3}{5}, -\frac{4}{25}, \frac{5}{125}, -\frac{6}{625}, \frac{7}{3125}, \dots \right\}$$

- **Definition**

A sequence $\{a_n\}$ is said to have limit L or **converge** to L , denoted by

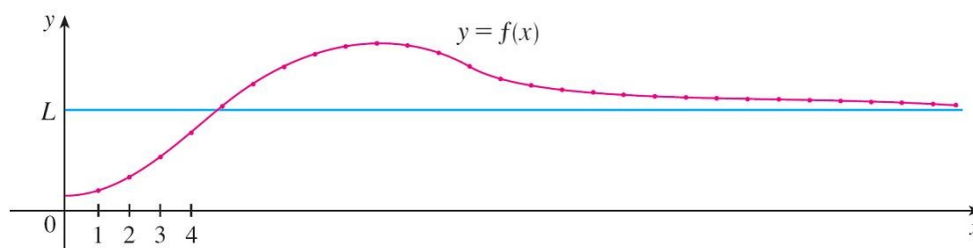
$$\lim_{n \rightarrow \infty} a_n = L$$

if the values of a_n get arbitrary close to L when n is large enough.



- **Theorem**

If $\lim_{x \rightarrow \infty} f(x) = L$, then $\lim_{n \rightarrow \infty} f(n) = L$.



- **Example** Determine whether the sequence $a_n = \frac{1}{\sqrt{n}}$ converges or diverges. If it converges, find the limit.

- **Example** Determine whether the sequence $\{(-1)^n\}$ converges or diverges. If it converges, find the limit.

- **Properties of Limits of Sequences**

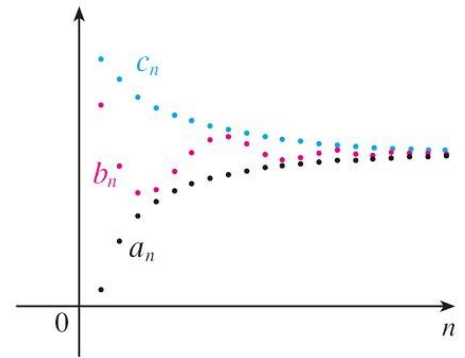
Let $\{a_n\}$ and $\{b_n\}$ be convergent sequences and k a constant. Then

1. $\lim_{n \rightarrow \infty} k = k$
2. $\lim_{n \rightarrow \infty} ka_n = k \lim_{n \rightarrow \infty} a_n$
3. $\lim_{n \rightarrow \infty} (a_n \pm b_n) = \lim_{n \rightarrow \infty} a_n \pm \lim_{n \rightarrow \infty} b_n$
4. $\lim_{n \rightarrow \infty} (a_n \cdot b_n) = \lim_{n \rightarrow \infty} a_n \cdot \lim_{n \rightarrow \infty} b_n$
5. $\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \frac{\lim_{n \rightarrow \infty} a_n}{\lim_{n \rightarrow \infty} b_n}$, if $\lim_{n \rightarrow \infty} b_n \neq 0$.

- **Example** Find $\lim_{n \rightarrow \infty} \frac{3n^2}{7n^2 + 1}$.

- **Example** Calculate $\lim_{n \rightarrow \infty} \frac{\ln n}{n}$.

- **Example** Show that if $\lim_{n \rightarrow \infty} |a_n| = 0$, then $\lim_{n \rightarrow \infty} a_n = 0$.



- **Example** Evaluate $\lim_{n \rightarrow \infty} \frac{(-1)^n}{n}$ if it exists.

- **Theorem**

If $\lim_{n \rightarrow \infty} a_n = L$ and the function f is continuous at L , then

$$\lim_{n \rightarrow \infty} f(a_n) = f(L)$$

- **Example** Find $\lim_{n \rightarrow \infty} \sin\left(\frac{\pi}{n}\right)$.

- **Definition**

- A sequence $\{a_n\}$ is called **increasing** if $a_n < a_{n+1}$ for all n .
- A sequence $\{a_n\}$ is called **decreasing** if $a_n > a_{n+1}$ for all n .
- A sequence that is increasing or decreasing is called a **monotonic sequence**.

- **Definition**

- A sequence $\{a_n\}$ is **bounded above** if there is a number M such that

$$a_n \leq M \quad \forall n \geq 1$$

- A sequence $\{a_n\}$ is **bounded below** if there is a number m such that

$$a_n \geq m \quad \forall n \geq 1$$

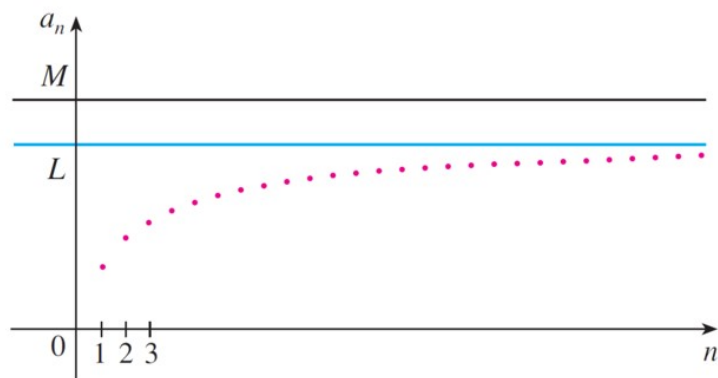
- If a sequence is bounded above and bounded below, we say that it is a **bounded sequence**.

- **Example** Show that the sequence $\left\{ \frac{n}{n+2} \right\}$ is increasing. Is the sequence bounded?

- **Example** Show that the sequence $\left\{ \frac{n}{n^2+1} \right\}$ is decreasing. Is the sequence bounded?

- **Monotonic Sequence Theorem**

Every bounded, monotonic sequence is convergent.



- **Example** Suppose you know that $\{a_n\}$ is an increasing sequence and all its terms lie between the numbers $\frac{1}{3}$ and 1. Explain why the sequence has a limit? What can you say about the value of the limit?

Exercises 11.1 3, 5, 9, 11, 13, 15, 17, 19, 21, 23, 33, 53, 55, 59.

11.2 Series

- An **infinite series** is of the form

$$a_1 + a_2 + a_3 + a_4 + \cdots$$

which is also denoted by $\sum_{n=1}^{\infty} a_n$ or $\sum a_n$.

- Example** $\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \cdots = ?$

n	Sum of first n terms
1	0.50000000
2	0.75000000
3	0.87500000
4	0.93750000
5	0.96875000
6	0.98437500
7	0.99218750
10	0.99902344
15	0.99996948
20	0.99999905
25	0.99999997

- The n th **partial sum** of an infinite series $\sum a_n$, denoted by S_n , is given by

$$S_n = a_1 + a_2 + \cdots + a_n = \sum_{i=1}^n a_i$$

- Example** Show that the n th partial sum S_n of a geometric series $a + ar + ar^2 + ar^3 + \cdots$ is $\frac{a(1 - r^n)}{1 - r}$.

- **Definition**

The infinite series $\sum_{n=1}^{\infty} a_n$ converges and has sum S if the sequence of partial sums $\{S_n\}$ converges to S . If $\{S_n\}$ diverges, then the series diverges. A divergent series has no sum.

- **Example** Show that a geometric series $a + ar + ar^2 + ar^3 + \cdots$ converges, and has sum $S = \frac{a}{1-r}$ if $-1 < r < 1$.

- **Example** Find the sum of the series

$$5 - \frac{10}{3} + \frac{20}{9} - \frac{40}{27} + \cdots$$

- **Example** Show that the series $\sum_{n=1}^{\infty} \frac{1}{4n^2 - 1}$ is convergent by expressing S_n as a **telescoping sum** and find its sum.

- ***n*th-Term Test for Divergence**

If the series $\sum_{n=1}^{\infty} a_n$ converges, then $\lim_{n \rightarrow \infty} a_n = 0$.

Equivalently, if $\lim_{n \rightarrow \infty} a_n \neq 0$ or if $\lim_{n \rightarrow \infty} a_n$ does not exist, then the series diverges.

- **Example** Determine whether the series $\sum_{n=1}^{\infty} \frac{n^3}{3n^3 + 2n^2}$ is convergent or divergent.

- **Example** Show that the **harmonic series**

$$\sum_{n=1}^{\infty} \frac{1}{n} = 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \cdots$$

is divergent.

- **Properties of Convergent Series**

If $\sum a_n$ and $\sum b_n$ are convergent series, then so are the series $\sum ca_n$ (where c is a constant) and $\sum (a_n + b_n)$, and $\sum (a_n - b_n)$, and

$$(i) \sum_{n=1}^{\infty} (ca_n) = c \sum_{n=1}^{\infty} a_n$$

$$(ii) \sum_{n=1}^{\infty} (a_n + b_n) = \sum_{n=1}^{\infty} a_n + \sum_{n=1}^{\infty} b_n$$

$$(iii) \sum_{n=1}^{\infty} (a_n - b_n) = \sum_{n=1}^{\infty} a_n - \sum_{n=1}^{\infty} b_n$$

Conversely, if $\sum_{n=1}^{\infty} a_n$ diverges and $c \neq 0$, then $\sum_{n=1}^{\infty} ca_n$ also diverges.

- **Example** Find the sum of the series $\sum_{n=1}^{\infty} \left[\frac{1}{4n^2 - 1} - 5 \left(\frac{1}{3} \right)^n \right]$.

- **Example** Determine whether the series $\sum_{n=1}^{\infty} \frac{1}{3n}$ is convergent or divergent.

Exercises 11.2 9, 15, 17, 19, 21, 23, 27.

11.3 The Integral Test

• Integral Test

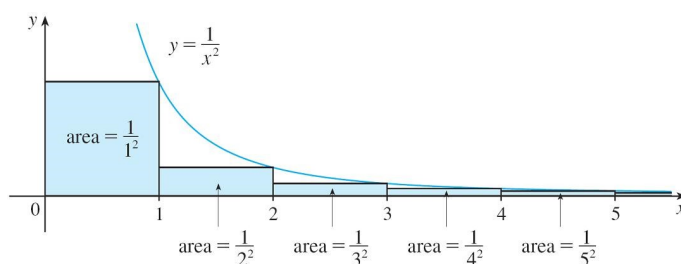
Let f be a continuous, **positive**, decreasing function on the interval $[1, \infty)$ and suppose that $a_n = f(n)$ for all positive integer n . Then the infinite series

$$\sum_{n=1}^{\infty} a_n$$

converges if and only if the improper integral

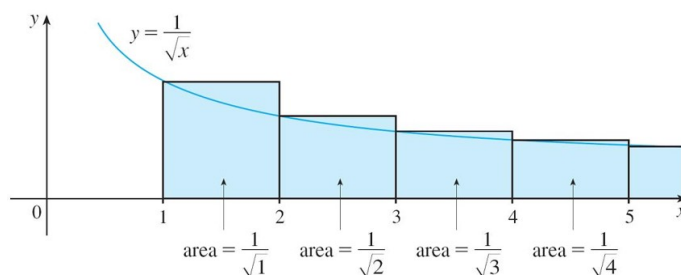
$$\int_1^{\infty} f(x) dx$$

converges.



$$\bullet \sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \frac{1}{4^2} + \frac{1}{5^2} + \cdots$$

$$\bullet \int_1^{\infty} \frac{1}{x^2} dx$$



$$\bullet \sum_{n=1}^{\infty} \frac{1}{\sqrt{n}} = \frac{1}{\sqrt{1}} + \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{3}} + \frac{1}{\sqrt{4}} + \frac{1}{\sqrt{5}} + \cdots$$

$$\bullet \int_1^{\infty} \frac{1}{\sqrt{x}} dx$$

- If p is a constant, the series

$$\sum_{n=1}^{\infty} \frac{1}{n^p}$$

is called a p -series.

- **Example** Show that the p -series is convergent if $p > 1$.

- **Example** Determine whether the series $\sum_{n=4}^{\infty} \frac{1}{n^{1.001}}$ is convergent or divergent.

- **Example** Determine whether the series $\sum_{n=1}^{\infty} \frac{\ln n}{n}$ is convergent or divergent.

Exercises 11.3 5, 7, 9, 13, 19, 21.

11.4 The Comparison Tests

- Ordinary Comparison Test

Suppose that $\sum a_n$ and $\sum b_n$ are series with **positive terms**.

(i) If $\sum b_n$ is convergent and $a_n \leq b_n$ for all n , then $\sum a_n$ is also convergent.

(ii) If $\sum b_n$ diverges and $a_n \geq b_n$ for all n , then $\sum a_n$ is also divergent.

- **Example** Determine whether the series $\sum_{n=1}^{\infty} \frac{n}{5n^2 - 4}$ is convergent or divergent.

- **Example** Determine whether the series $\sum_{n=1}^{\infty} \frac{n}{2^n(n+1)}$ is convergent or divergent.

- **Limit Comparison Test**

Suppose that $\sum a_n$ and $\sum b_n$ are series with **positive terms**. If

$$\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = c$$

where c is a finite number and $c > 0$, then either both series converge or both diverge.

- **Example** Determine the convergence or divergence of each series.

(a) $\sum_{n=1}^{\infty} \frac{3n-2}{n^3-2n^2+11}$

(b) $\sum_{n=1}^{\infty} \frac{1}{\sqrt{n^2+19}}$

Exercises 11.4 3, 9, 11, 19, 21, 23, 25.

11.5 Alternating Series

- An alternating series is a series of the form

$$a_1 - a_2 + a_3 - a_4 + a_5 - a_6 + \dots$$

where $a_n \geq 0$.

- Alternating Series Test**

If the alternating series

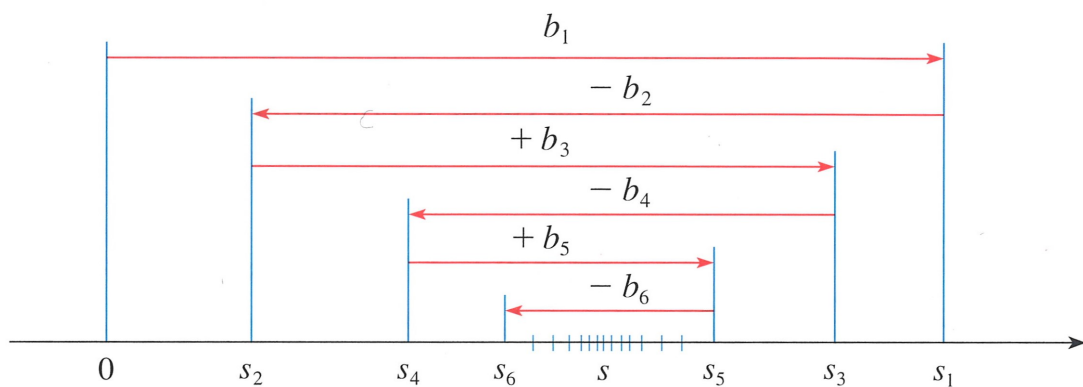
$$\sum_{n=1}^{\infty} (-1)^{n-1} b^n = b_1 - b_2 + b_3 - b_4 + b_5 - b_6 + \dots \quad b_n > 0$$

satisfies

(i) $b_n \geq b_{n+1}$ for all n

(ii) $\lim_{n \rightarrow \infty} b_n = 0$

then the series is convergent.



- **Example** Determine whether the alternating harmonic series

$$1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} - \frac{1}{6} + \cdots$$

is convergent or divergent.

- **Example** Show that the series $\frac{1}{1!} - \frac{1}{2!} + \frac{1}{3!} - \frac{1}{4!} + \cdots$ is convergent.

- **Example** Test the series for convergence or divergence

$$\sum_{n=1}^{\infty} \frac{(-1)^n 3n}{4n - 1}$$

Exercises 11.5 5, 7, 9, 13.

11.6 Absolute Convergence and the Ratio Test

- **Definition**

- A series $\sum a_n$ is called **absolutely convergent** if the series of absolute values $\sum |a_n|$ is convergent.
- A series $\sum a_n$ is called **conditionally convergent** if it is convergent but not absolutely convergent.

- **Absolute Convergence Test**

If $\sum |a_n|$ converges, then $\sum a_n$ converges.

- **Example** Show that $\sum_{n=1}^{\infty} (-1)^{n-1} \frac{1}{n^2}$ converges.

- **Example** Show that $\sum_{n=1}^{\infty} (-1)^{n+1} \frac{1}{\sqrt{n}}$ is conditionally convergent.

- **Absolute Ratio Test**

Suppose that

$$\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = L$$

- (i) If $L < 1$, the series converges absolutely.
- (ii) If $L > 1$, the series diverges.
- (iii) If $L = 1$, the test is inconclusive.

- **Example** Show that $\sum_{n=1}^{\infty} (-1)^{n+1} \frac{3^n}{n!}$ converges absolutely.

11.8 Power Series

- A power series in x is a series of the form

$$\sum_{n=0}^{\infty} c_n x^n = c_0 + c_1 x + c_2 x^2 + c_3 x^3 + c_4 x^4 + \dots$$

where x is a **variable** and the c_n 's are constants called the **coefficients** of the series.

- **Example** For what x does the power series

$$\sum_{n=0}^{\infty} x^n = 1 + x + x^2 + x^3 + x^4 + \dots$$

converge and what is its sum?

- **Interval of Convergence** We call the set on which a power series converges its interval of convergence.

- **Example** Find the interval of convergence for $\sum_{n=0}^{\infty} n!x^n$.

- **Example** Find the interval of convergence for $\sum_{n=0}^{\infty} \frac{x^n}{n!}$.

- **Example** Find the interval of convergence for

$$\sum_{n=0}^{\infty} \frac{x^n}{(n+1)2^n} = 1 + \frac{1}{2} \frac{x}{2} + \frac{1}{3} \frac{x^2}{2^2} + \frac{1}{4} \frac{x^3}{2^3} + \dots$$

- **Theorem**

The interval of convergence for a power series $\sum_{n=0}^{\infty} c_n x^n$ is always one of the following three types:

- (i) The single point $x = 0$.
- (ii) An interval $(-R, R)$, plus possibly one or both end points.
- (iii) The whole real line.

In each of these cases, the series is said to have **radius of convergence** 0, R and ∞ respectively.

- A series of the form

$$\sum_{n=0}^{\infty} c_n(x-a)^n = c_0 + c_1(x-a) + c_2(x-a)^2 + \dots$$

is called a **power series centered at a** or a **power series about a** .

- **Theorem**

The interval of convergence for a power series $\sum_{n=0}^{\infty} c_n(x-a)^n$ is always one of the following three types:

- (i) The single point $x = a$.
- (ii) An interval $(a - R, a + R)$, plus possibly one or both end points.
- (iii) The whole real line.

In each of these cases, the series is said to have **radius of convergence** 0, R and ∞ respectively.

- **Example** Find the interval of convergence and the radius of convergence for $\sum_{n=0}^{\infty} \frac{n(x+2)^n}{3^{n+1}}$.

Exercises 11.8 3, 7, 13, 15, 23.

11.9 Representations of Functions as Power Series

- **Example** Express $\frac{1}{1+x^2}$ as the sum of a power series and find the interval of convergence.

- **Example** Find a power series representation for $\frac{x^3}{x+2}$.

- **Theorem**

If the power series $\sum c_n(x-a)^n$ has radius of convergence $R > 0$, then the function f defined by

$$f(x) = c_0 + c_1(x-a) + c_2(x-a)^2 + \cdots = \sum_{n=0}^{\infty} c_n(x-a)^n$$

is differentiable (and therefore continuous) on the interval $(a-R, a+R)$ and

$$(i) \quad f'(x) = c_1 + 2c_2(x-a) + 3c_3(x-a)^2 + \cdots = \sum_{n=1}^{\infty} nc_n(x-a)^{n-1}$$

$$(ii) \quad \int f(x)dx = C + c_0(x-a) + c_1 \frac{(x-a)^2}{2} + c_2 \frac{(x-a)^3}{3} + \cdots$$

$$= C + \sum_{n=0}^{\infty} c_n \frac{(x-a)^{n+1}}{n+1}$$

The radii of convergence of the power series in Equations (i) and (ii) are both R .

- **Example** Determine the interval of convergence for

$$1 + 2x + 3x^2 + 4x^3 + \cdots$$

and find an explicit formula for this series when it is convergent.

- **Example** Determine the interval of convergence for

$$x + \frac{x^2}{2} + \frac{x^3}{3} + \frac{x^4}{4} + \dots$$

and find an explicit formula for this series when it is convergent.

- **Example** Find a power series representation for $\tan^{-1} x$.

Exercises 11.9 3, 9, 13, 15.

11.10 Taylor and Maclaurin Series

- **Theorem**

If f has a power series representation (expansion) at a , that is, if

$$f(x) = c_0 + c_1(x - a) + c_2(x - a)^2 + c_3(x - a)^3 + \cdots$$

for all $|x - a| < R$, then its coefficients are given by the formula

$$c_n = \frac{f^{(n)}(a)}{n!}$$

- The series

$$\sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!} (x - a)^n = f(a) + \frac{f'(a)}{1!} (x - a) + \frac{f''(a)}{2!} (x - a)^2 + \frac{f'''(a)}{3!} (x - a)^3 + \cdots$$

is called the **Taylor series** of the function f at a (or about a or centered at a).

If $a = 0$, the series is called the **Maclaurin series**.

- **n th-degree Taylor polynomial of f at a**

$$T_n(x) = \sum_{i=0}^n \frac{f^{(i)}(a)}{i!} (x-a)^i = f(a) + \frac{f'(a)}{1!} (x-a)^1 + \cdots + \frac{f^{(n)}(a)}{n!} (x-a)^n$$

- **Remainder**

$$R_n(x) = f(x) - T_n(x)$$

- **Taylor's Inequality**

If $|f^{(n+1)}(x)| \leq M$ for $|x-a| \leq d$, then the remainder $R_n(x)$ of the Taylor series satisfies the inequality

$$|R_n(x)| \leq \frac{M}{(n+1)!} |x-a|^{n+1} \quad \text{for } |x-a| \leq d$$

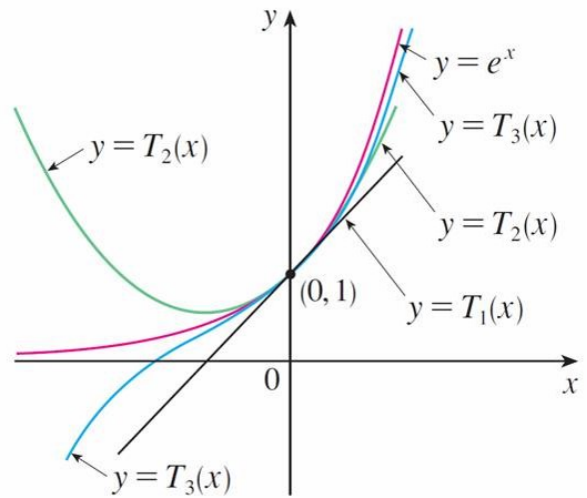
- **Theorem**

If $f(x) = T_n(x) + R_n(x)$ and

$$\lim_{n \rightarrow \infty} R_n(x) = 0$$

for $|x-a| < R$, then f is equal to the sum of its Taylor series on the interval $|x-a| < R$.

- **Example** Find the Maclaurin series of the function $f(x) = e^x$ and prove that it is equal to the sum of its Maclaurin series.



- **Example** Find the Taylor series for $f(x) = e^x$ at $a = 2$.

- **Example** Find the Maclaurin series for $\sin x$ and prove that it represents $\sin x$ for all x .
- **Example** Find the Maclaurin series for $\cos x$.
- **Example** Find the Maclaurin series for $x \cos x$.

- **Example** *Binomial Series* Find the Maclaurin series for $(1 + x)^k$ and show that it is convergent for $|x| < 1$.

- **Example** Obtain the Maclaurin series for $\frac{1}{(1-x)^2}$, $|x| < 1$.
- **Example** Find the Maclaurin series for the function $\sqrt{4+x}$ and determine its radius of convergence.

- **Important Maclaurin Series**

$$1. \frac{1}{1-x} = 1 + x + x^2 + x^3 + x^4 + \dots$$

$$2. \ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots$$

$$3. \tan^{-1} x = x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \dots$$

$$4. e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \dots$$

$$5. \sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots$$

$$6. \cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots$$

$$7. (1+x)^k = 1 + kx + \frac{k(k-1)}{2!}x^2 + \frac{k(k-1)(k-2)}{3!}x^3 + \dots$$

- **Example** Find the first three nonzero terms in the Maclaurin series for $e^x \sin x$.

- **Example** Find the first three nonzero terms in the Maclaurin series for $\tan x$.

Exercises 11.10 3, 7, 11, 13, 15, 19, 25, 27, 51, 53.

Exercises 11.11 1, 3, 11, 15.